```
Model Evaluation Report
Started: 2026-04-12 23:10:20
Finished: 2026-04-12 23:31:11

Model Architecture: ResNet50
Early Stopping Patience: 5
Phase 1 Learning Rate: 0.01
Phase 2 Learning Rate: 1e-06
Phase 1 Epochs:           18 (Limit: 30)
Phase 2 Epochs:           26 (Limit: 50)

Test Loss                : 0.5105
Binary Accuracy          : 0.9037
Precision                : 0.8719
Sensitivity (Recall)     : 0.9426
AUC                      : 0.9523
Specificity: 0.8660

Optimal Threshold found: 0.7428
Optimised sensitivity: 0.9426
Optimised specificity:0.8922

Classification Report:
              precision    recall  f1-score   support

      Normal       0.94      0.87      0.90       306
    Cataract       0.87      0.94      0.91       296

    accuracy                           0.90       602
   macro avg       0.91      0.90      0.90       602
weighted avg       0.91      0.90      0.90       602
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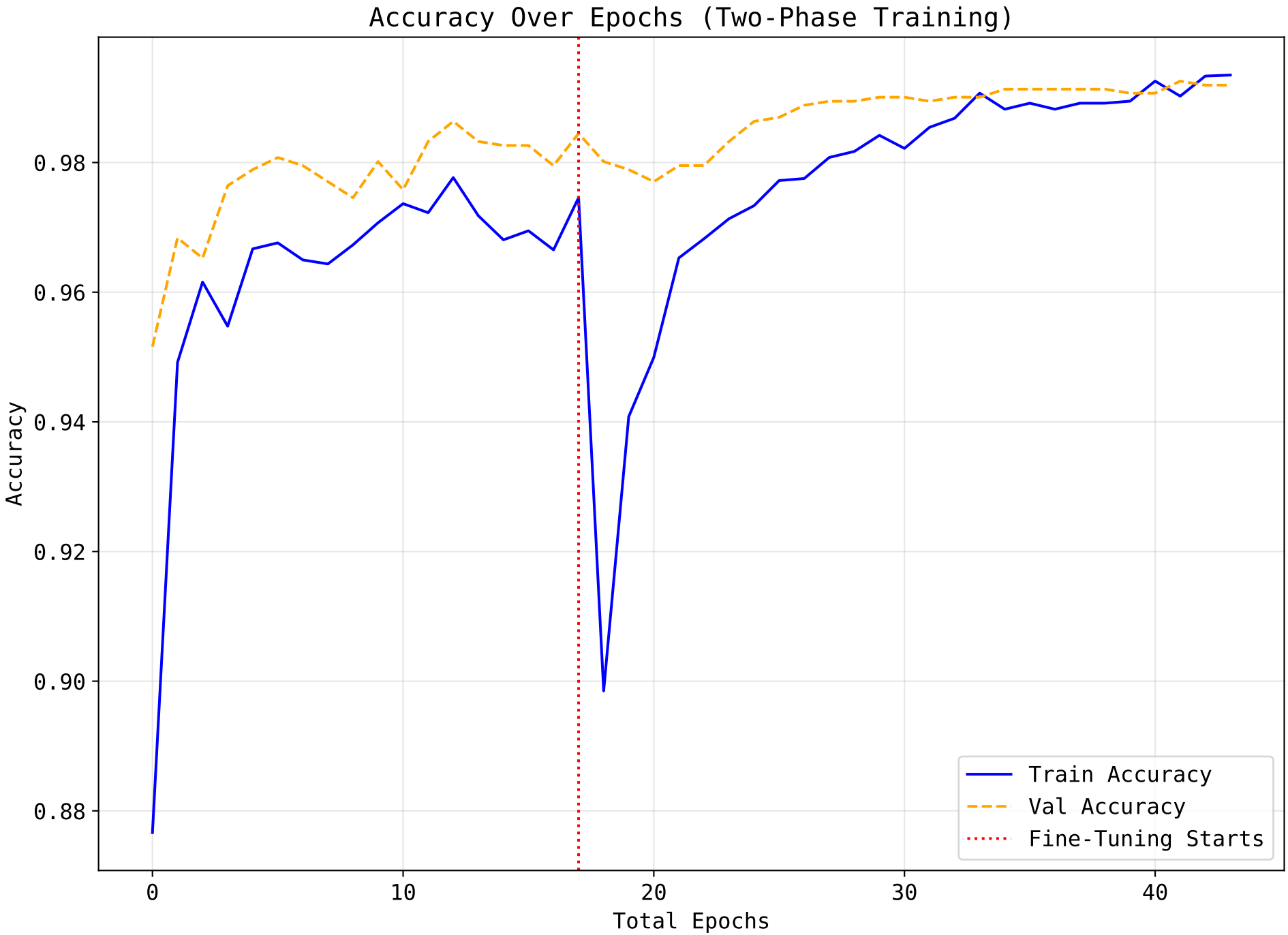
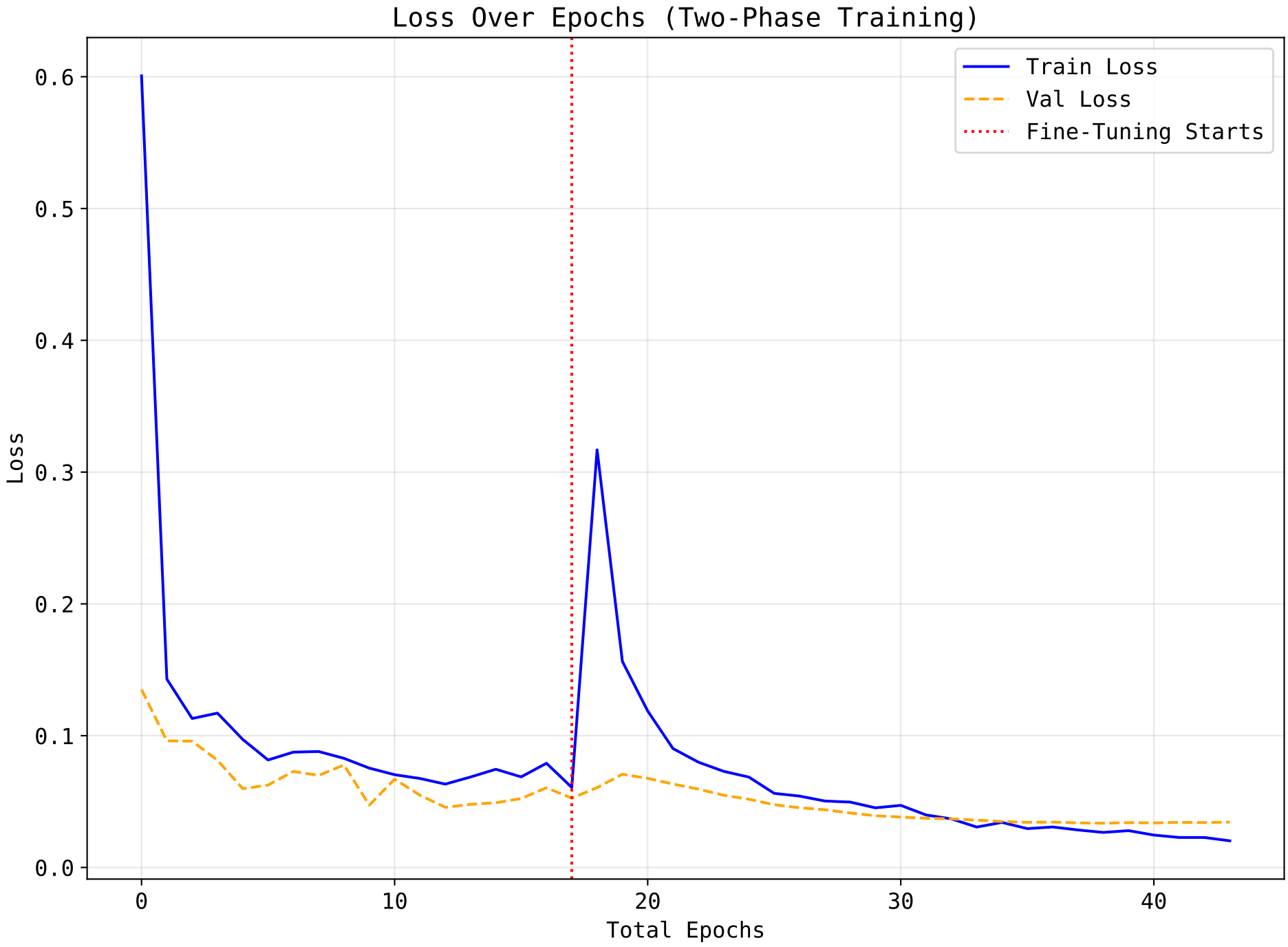
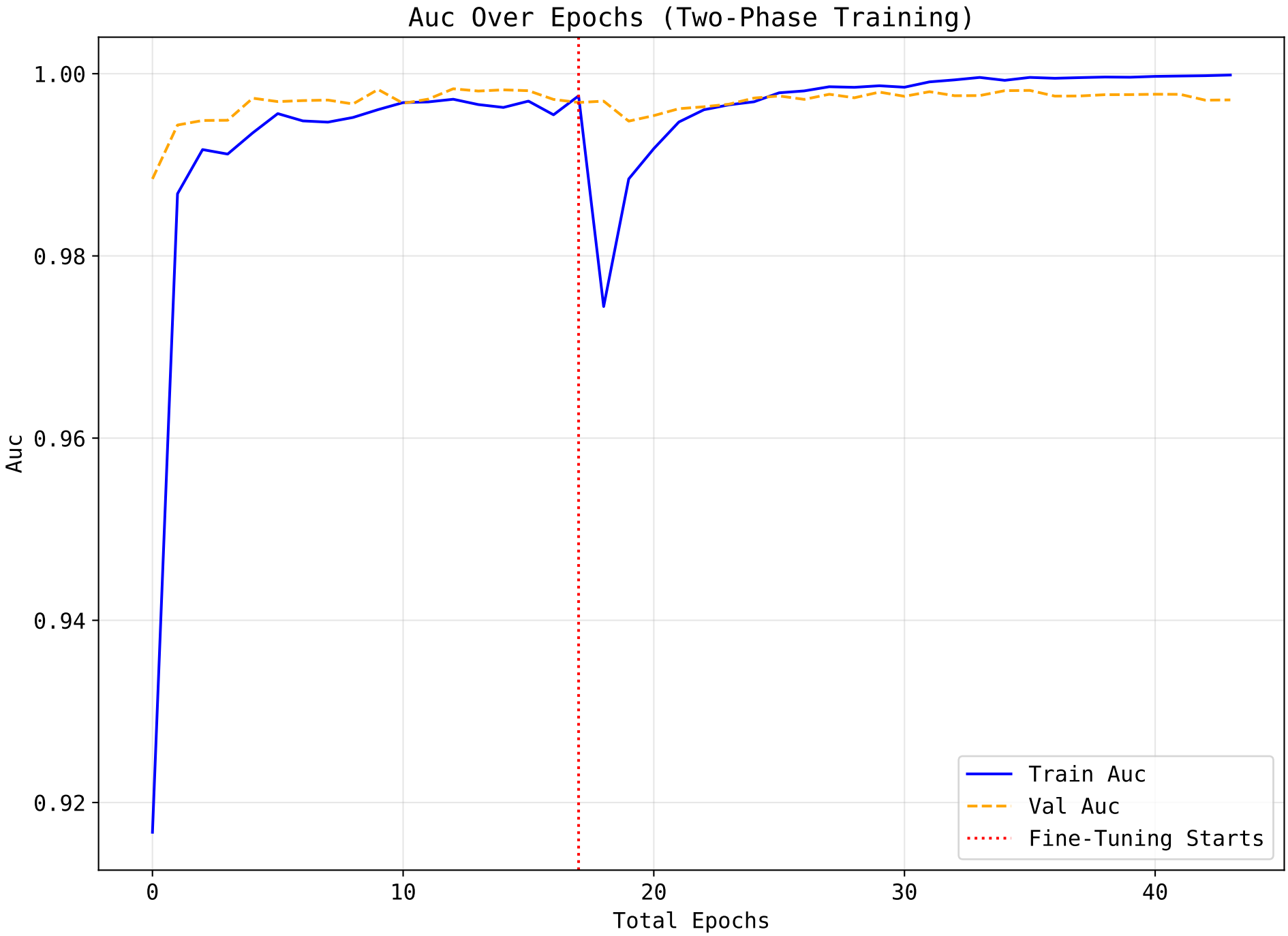
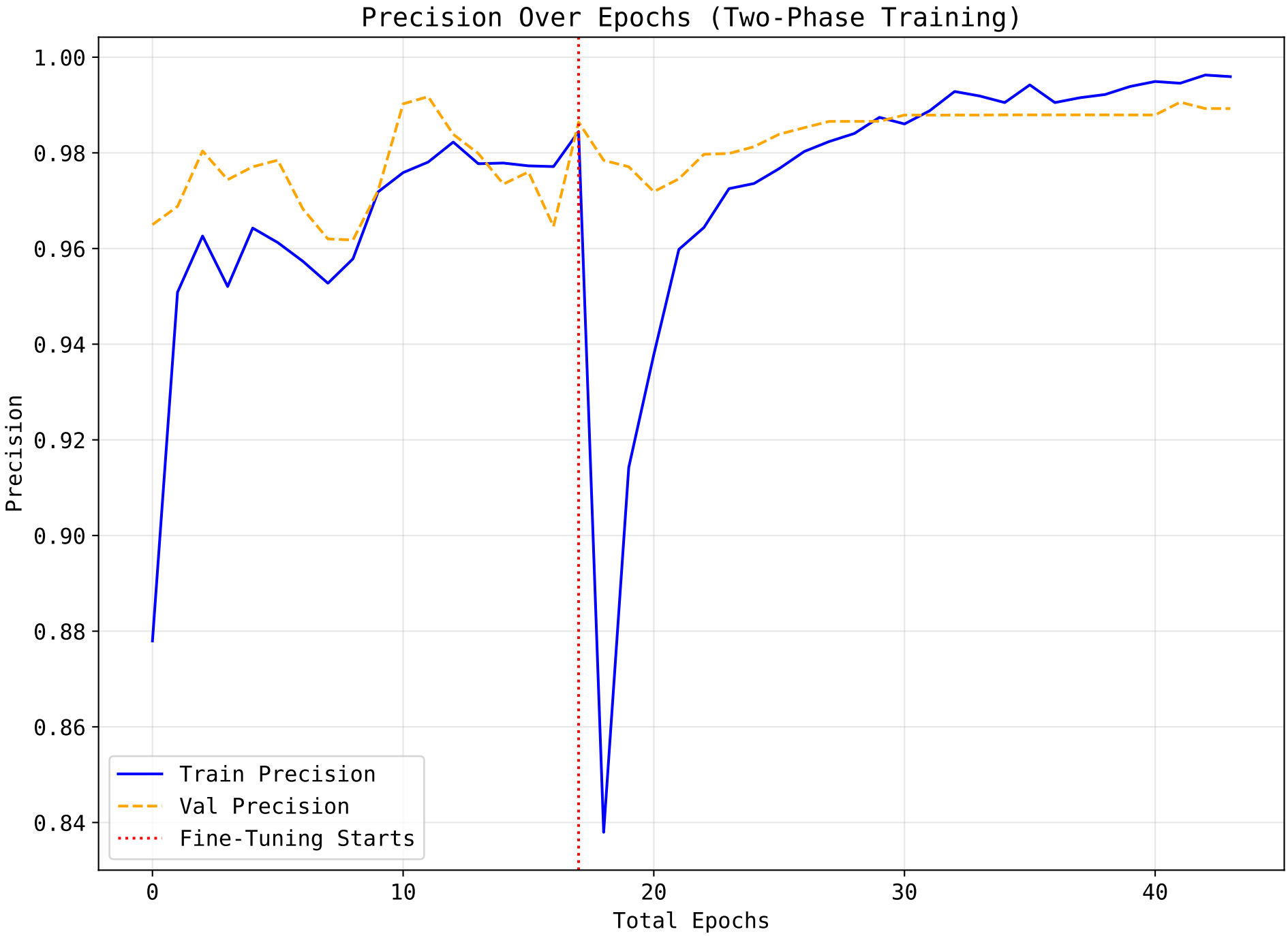
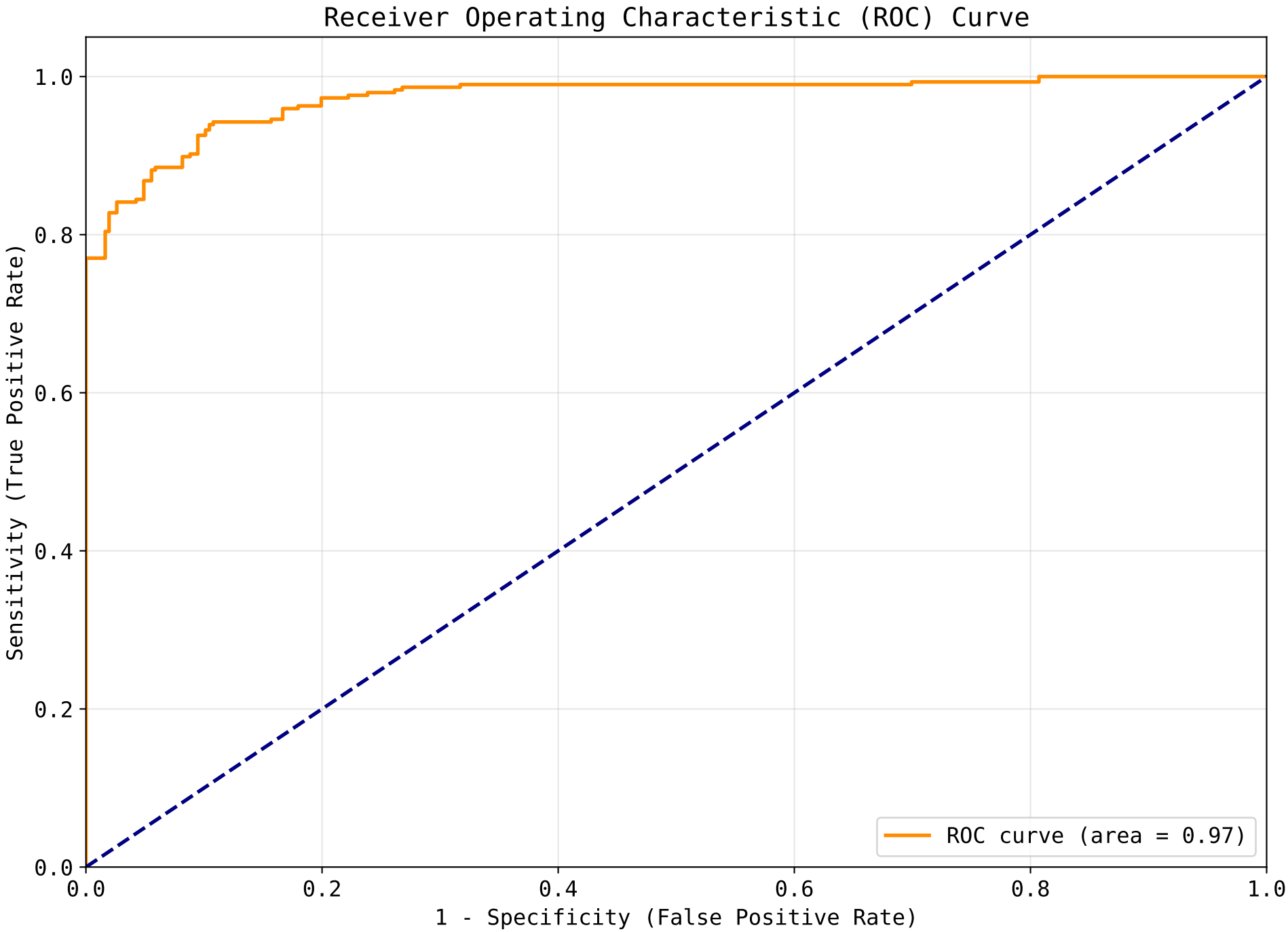

Accuracy Over Epochs (Two-Phase Training)
Accuracy
0.98
0.96
0.94
0.92
0.90
0.88
0
10
20
30
40
Total Epochs
Train Accuracy
Val Accuracy
Fine-Tuning Starts

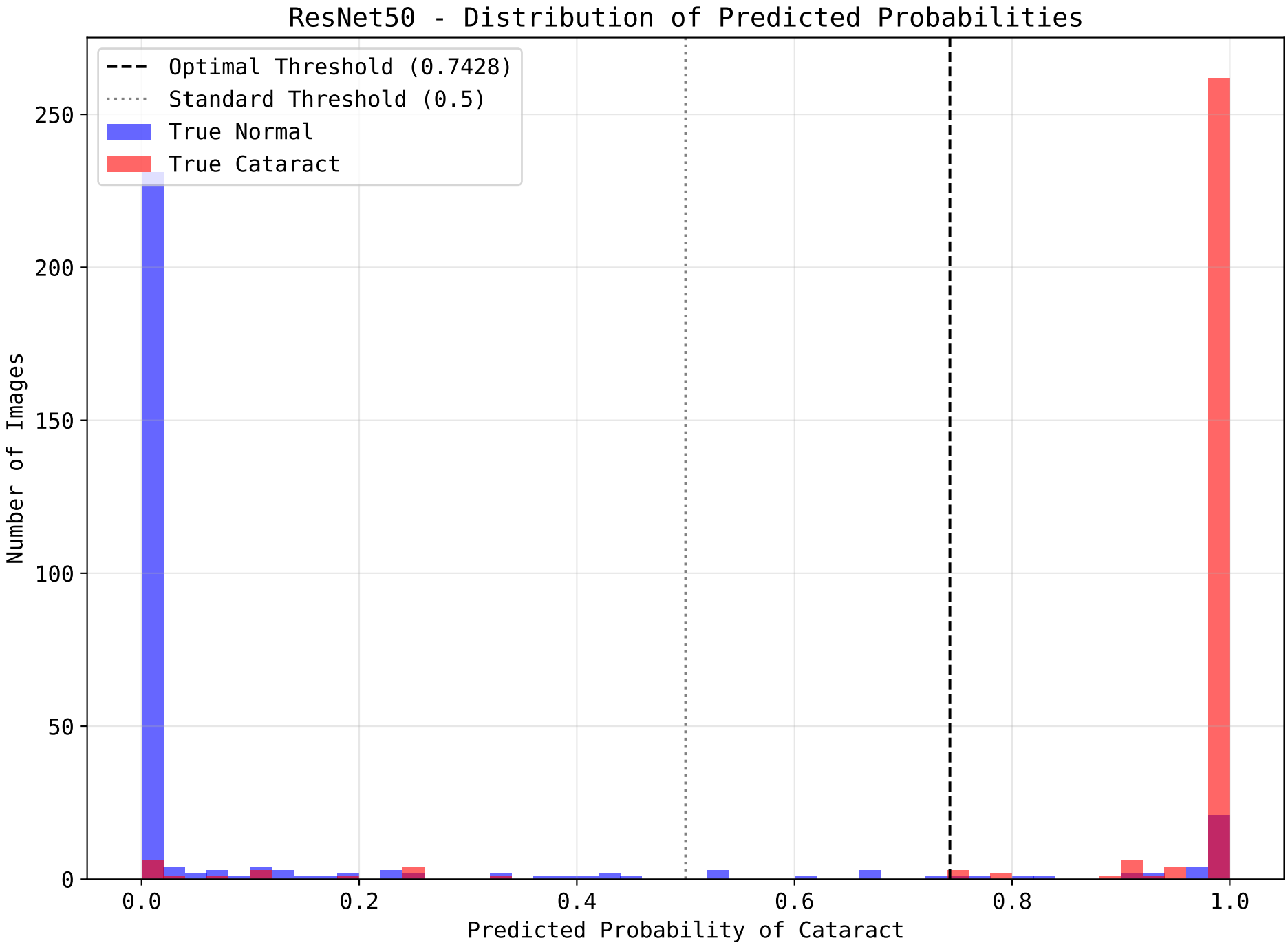

Loss Over Epochs (Two-Phase Training)
Train Loss
Val Loss
Fine-Tuning Starts
Loss
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
40
Total Epochs

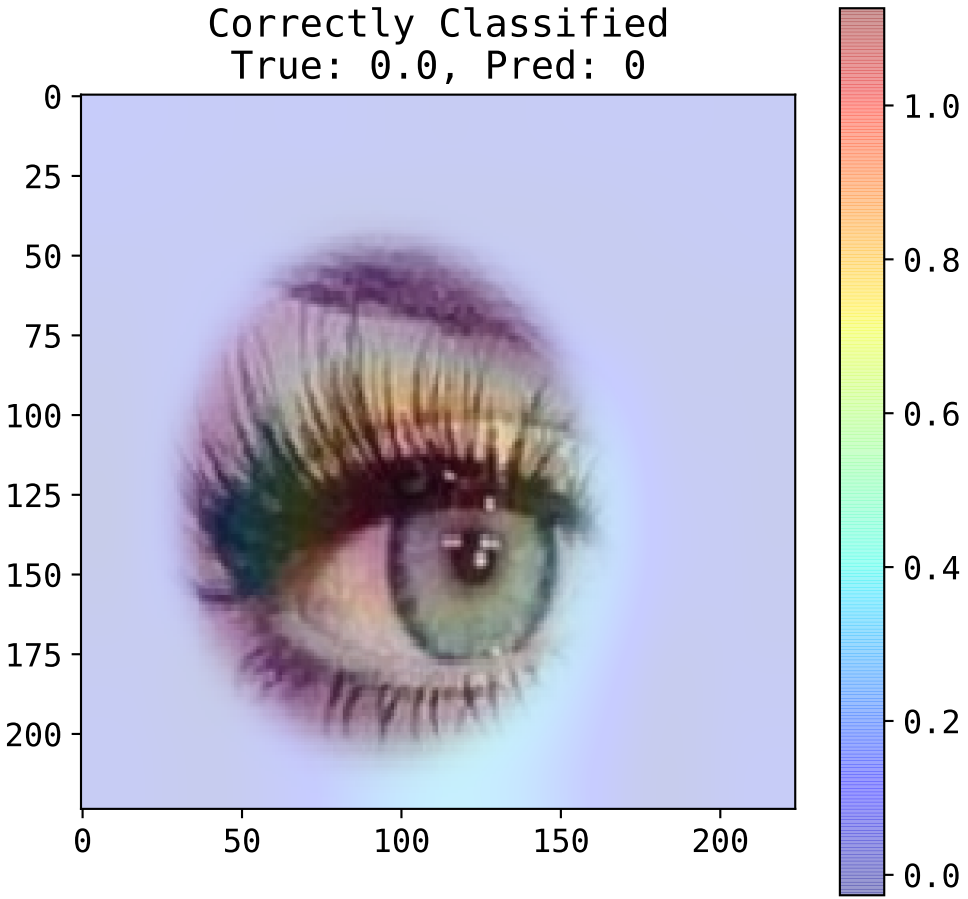
Sensitivity Over Epochs (Two-Phase Training)
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
Sensitivity
0
10
20
30
40
Total Epochs
Train Sensitivity
Val Sensitivity
Fine-Tuning Starts

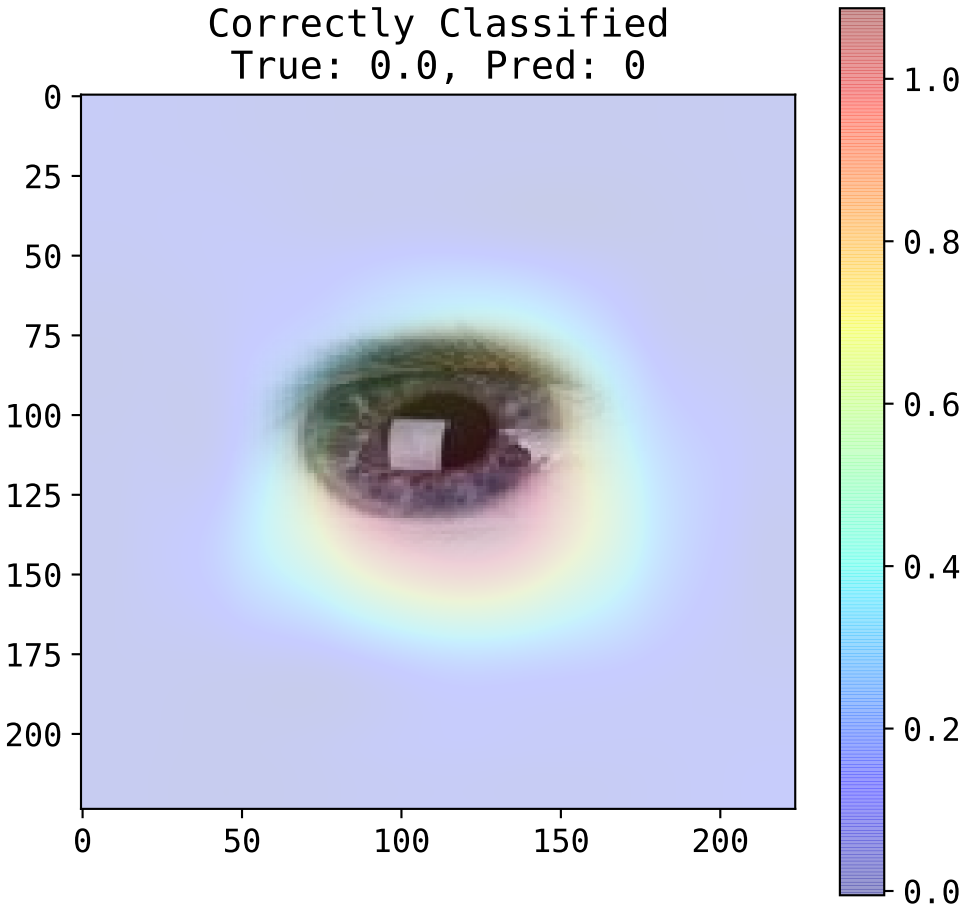

Auc Over Epochs (Two-Phase Training)
1.00
0.98
0.96
0.94
0.92
Auc
0
10
20
30
40
Total Epochs
Train Auc
Val Auc
Fine-Tuning Starts

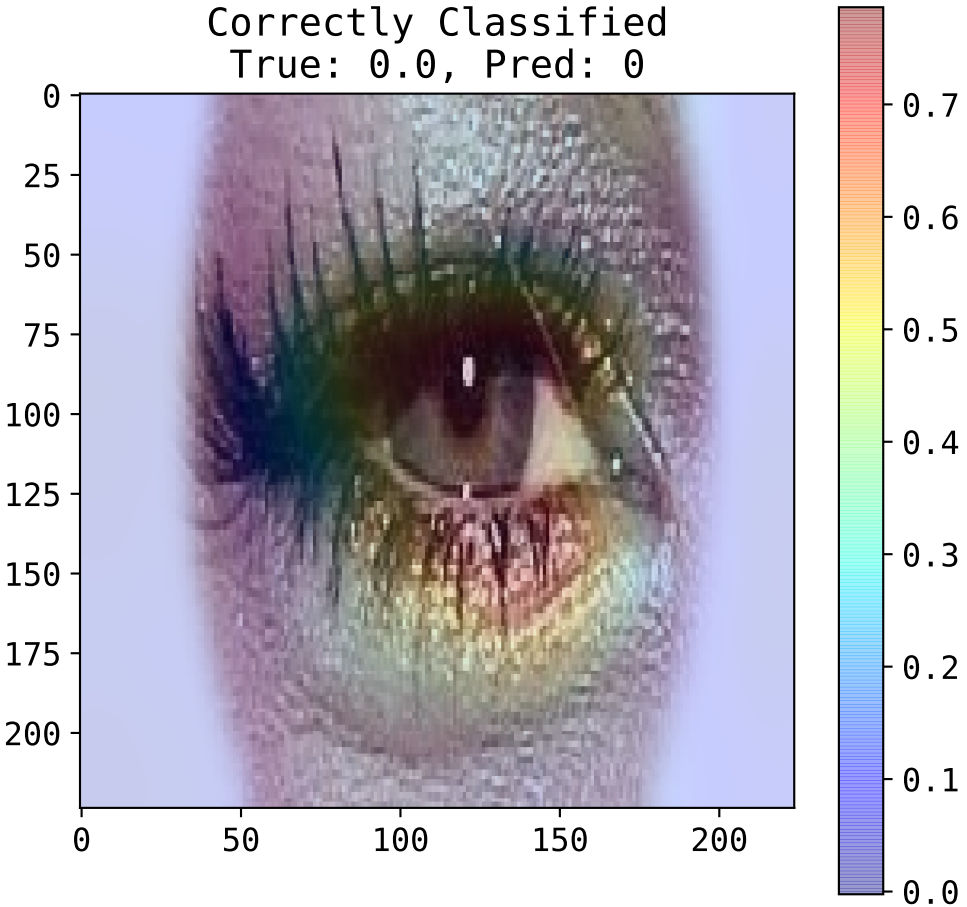

Precision Over Epochs (Two-Phase Training)
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.84
Precision
0
10
20
30
40
Total Epochs
Train Precision
Val Precision
Fine-Tuning Starts


Receiver Operating Characteristic (ROC) Curve
Sensitivity (True Positive Rate)
1 - Specificity (False Positive Rate)
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.97)


ResNet50 - Distribution of Predicted Probabilities
Optimal Threshold (0.7428)
Standard Threshold (0.5)
True Normal
True Cataract
Number of Images
0
50
100
150
200
250
0.0
0.2
0.4
0.6
0.8
1.0
Predicted Probability of Cataract


Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Correctly Classified
True: 0.0, Pred: 0
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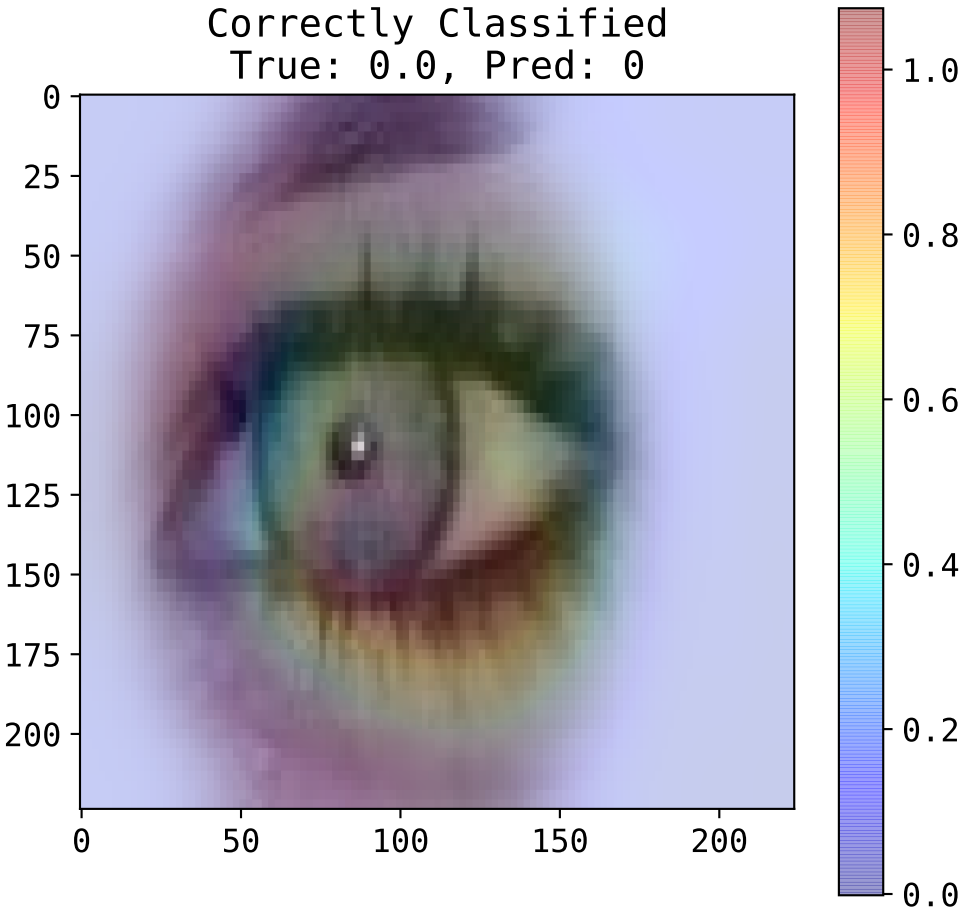
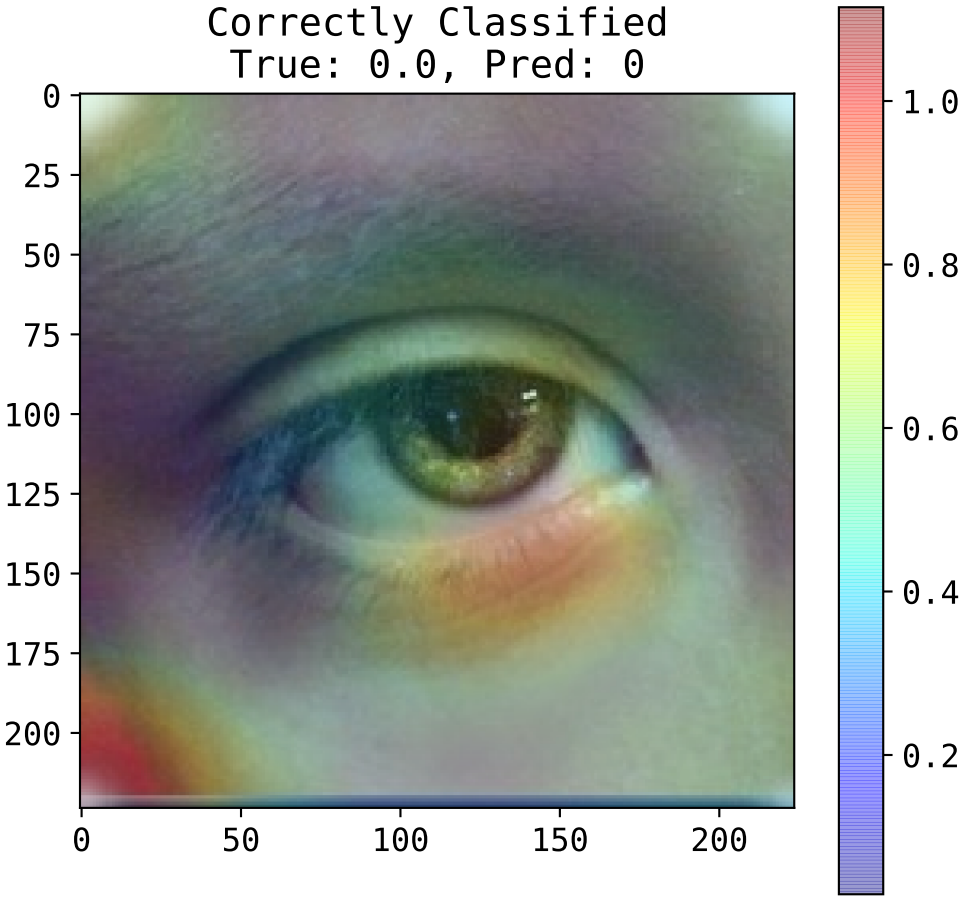
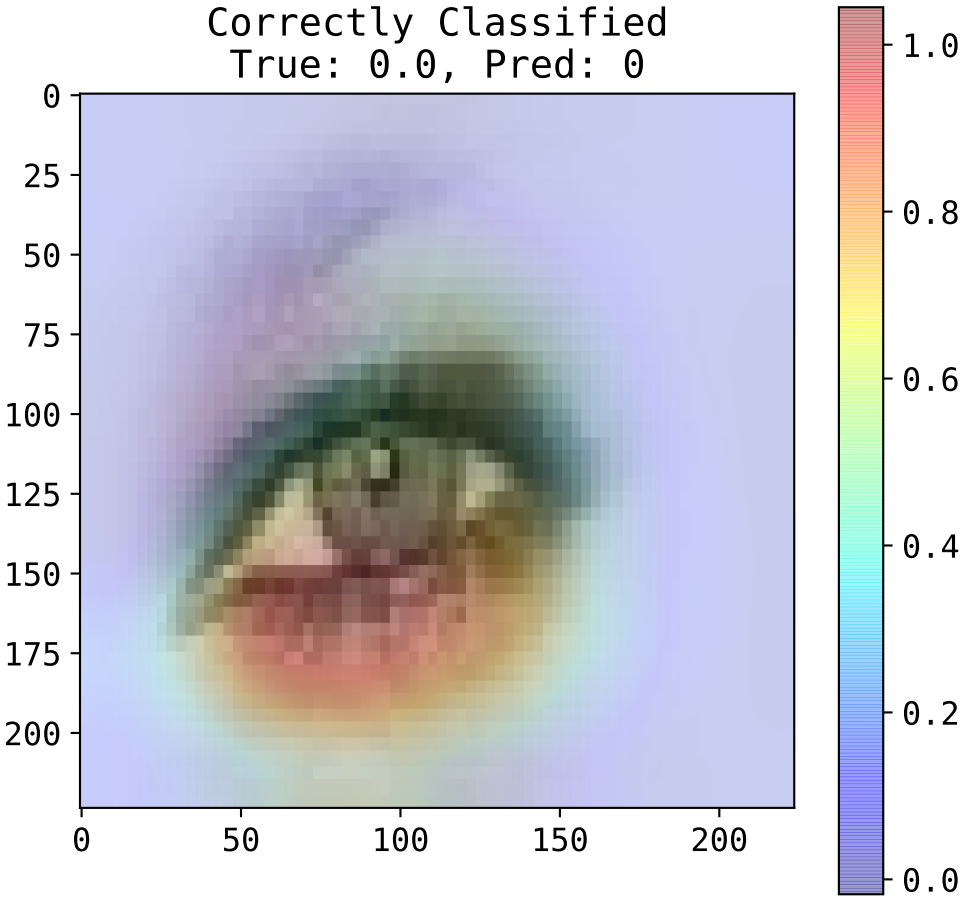
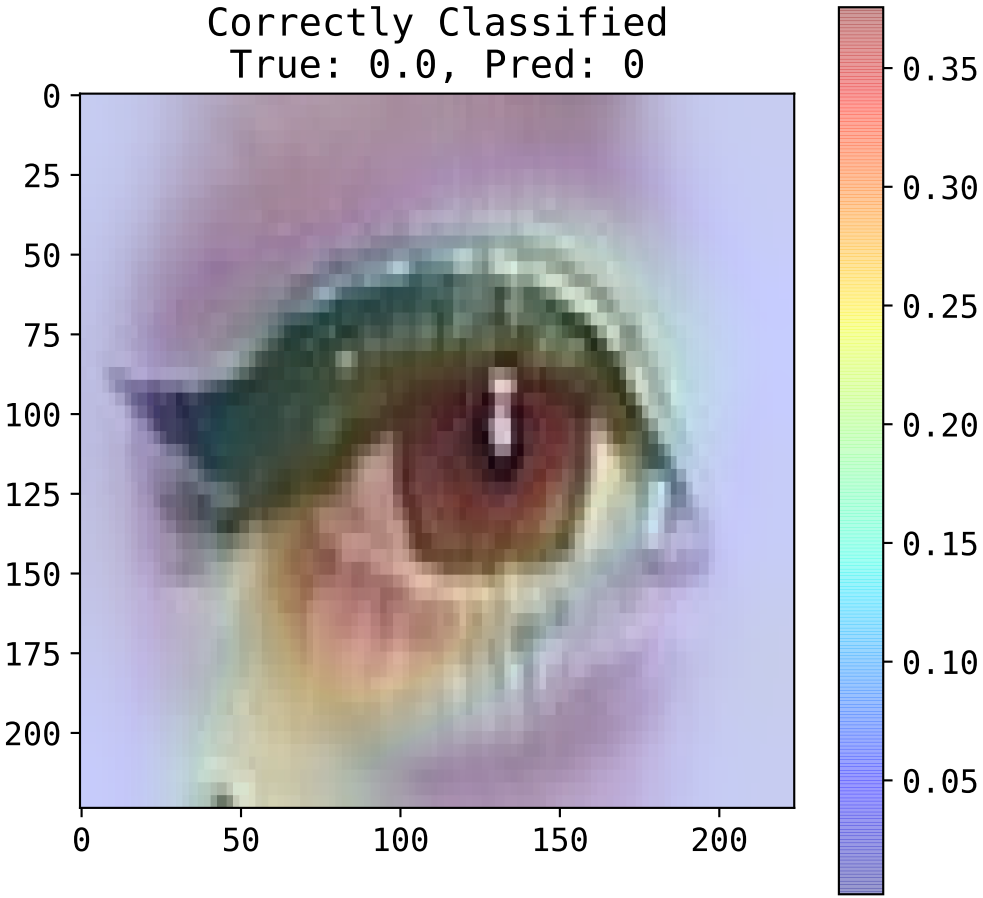
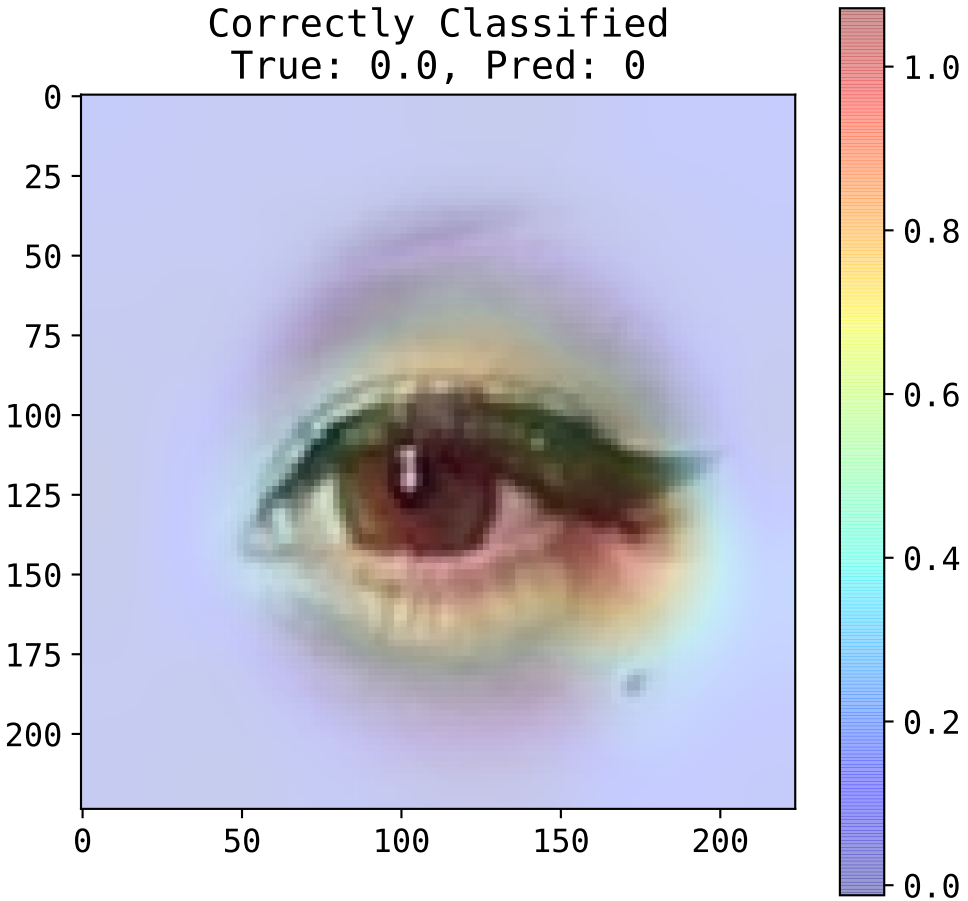
25
50
75
100
125
150
175
200
0
50
100
150
200
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

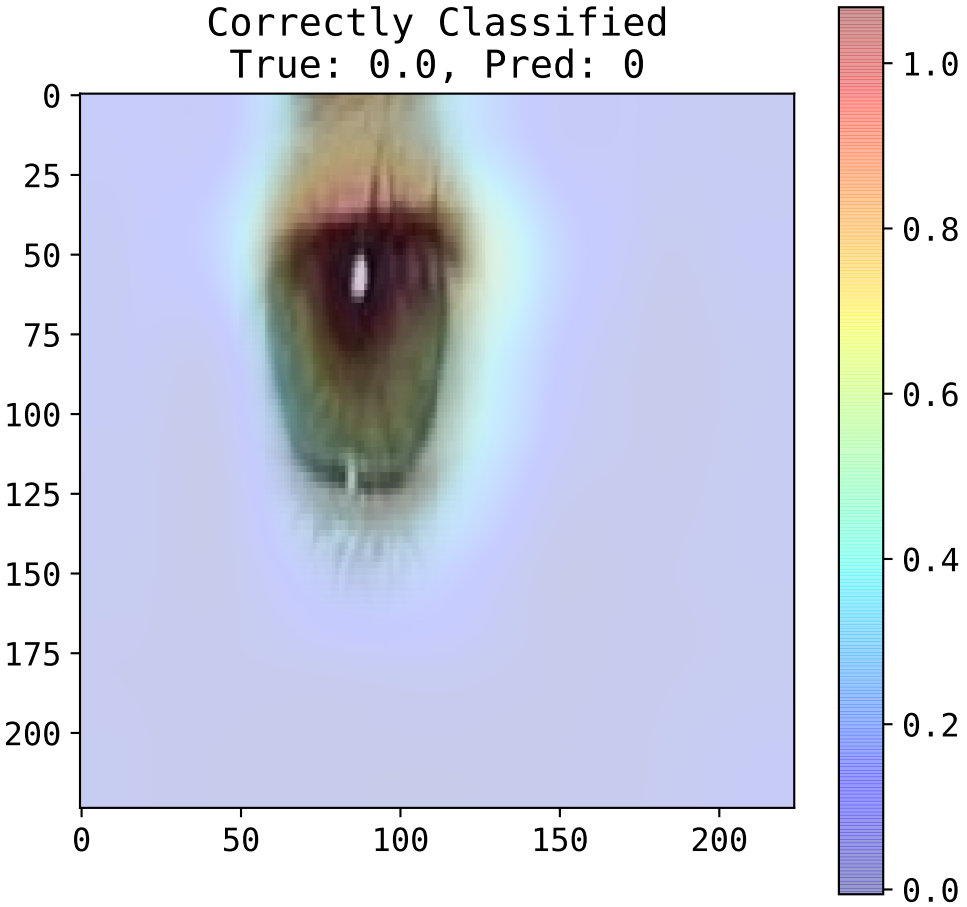

Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0

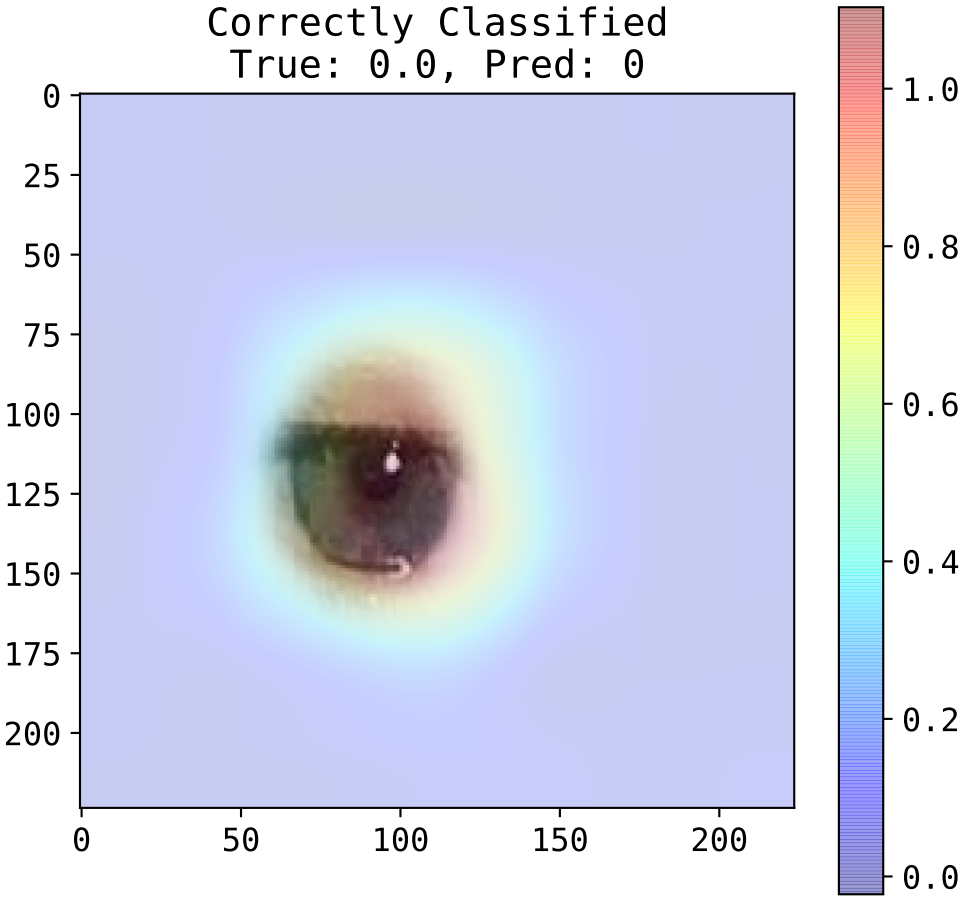

Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2

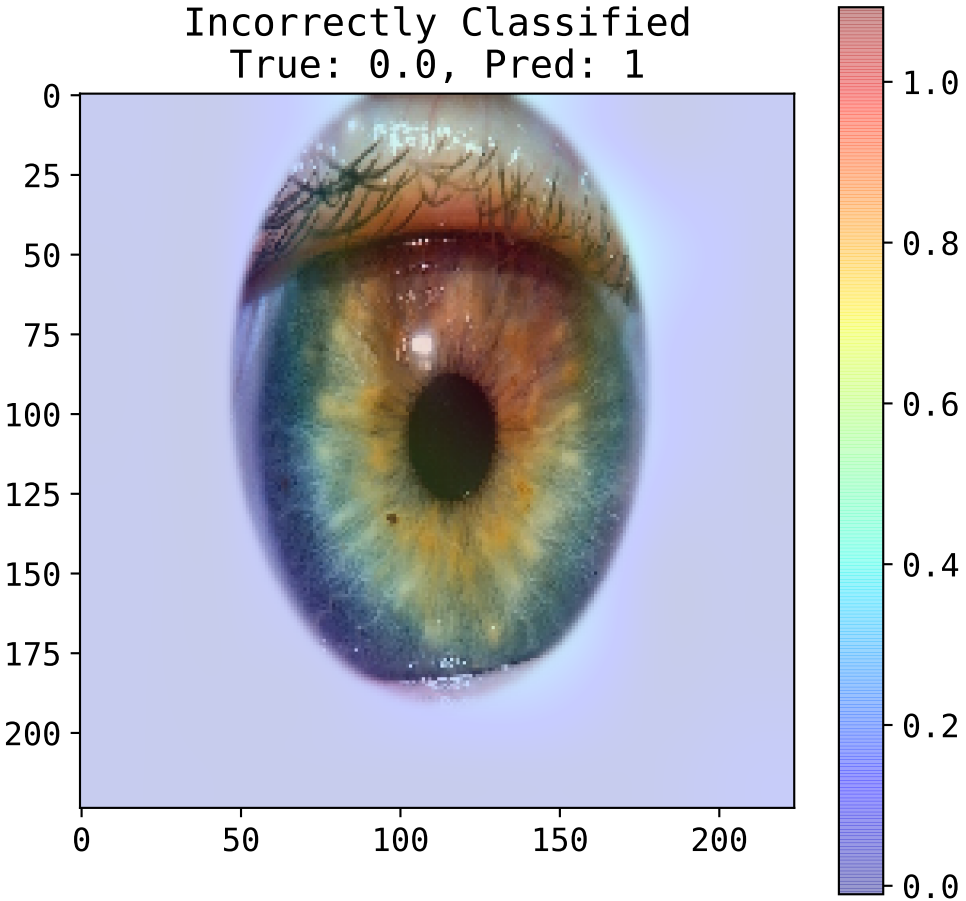

Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0

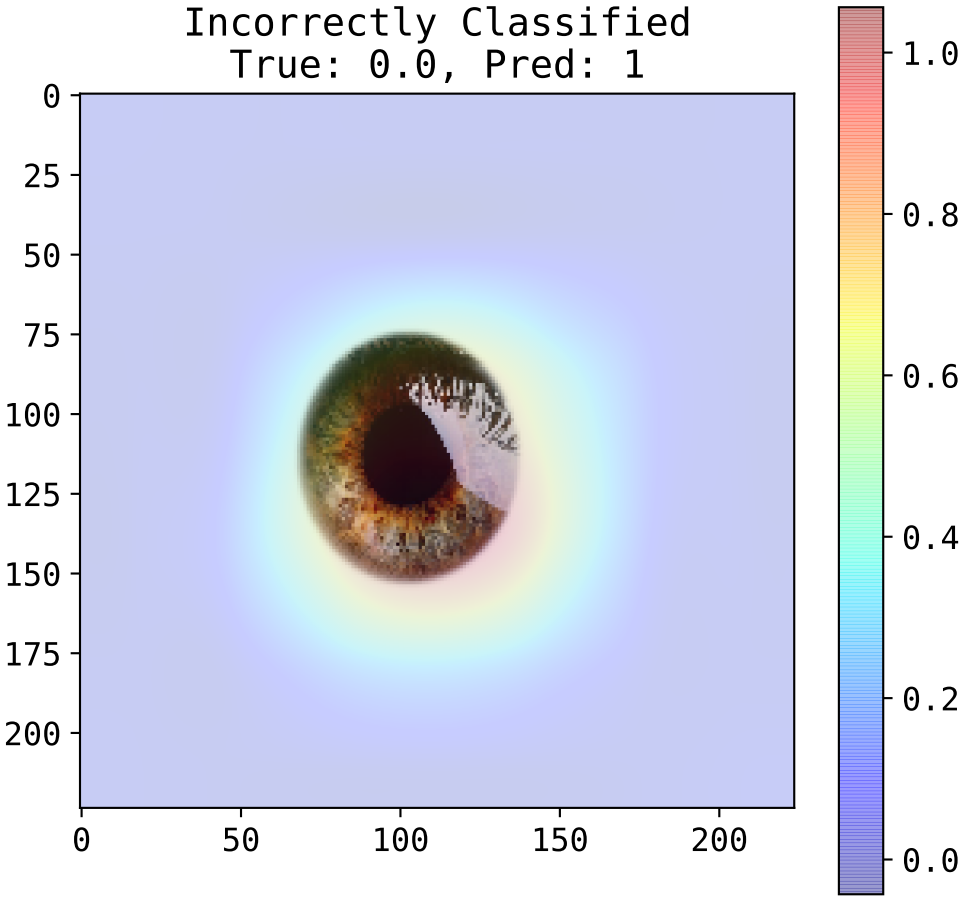

Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
0.35
0.30
0.25
0.20
0.15
0.10
0.05

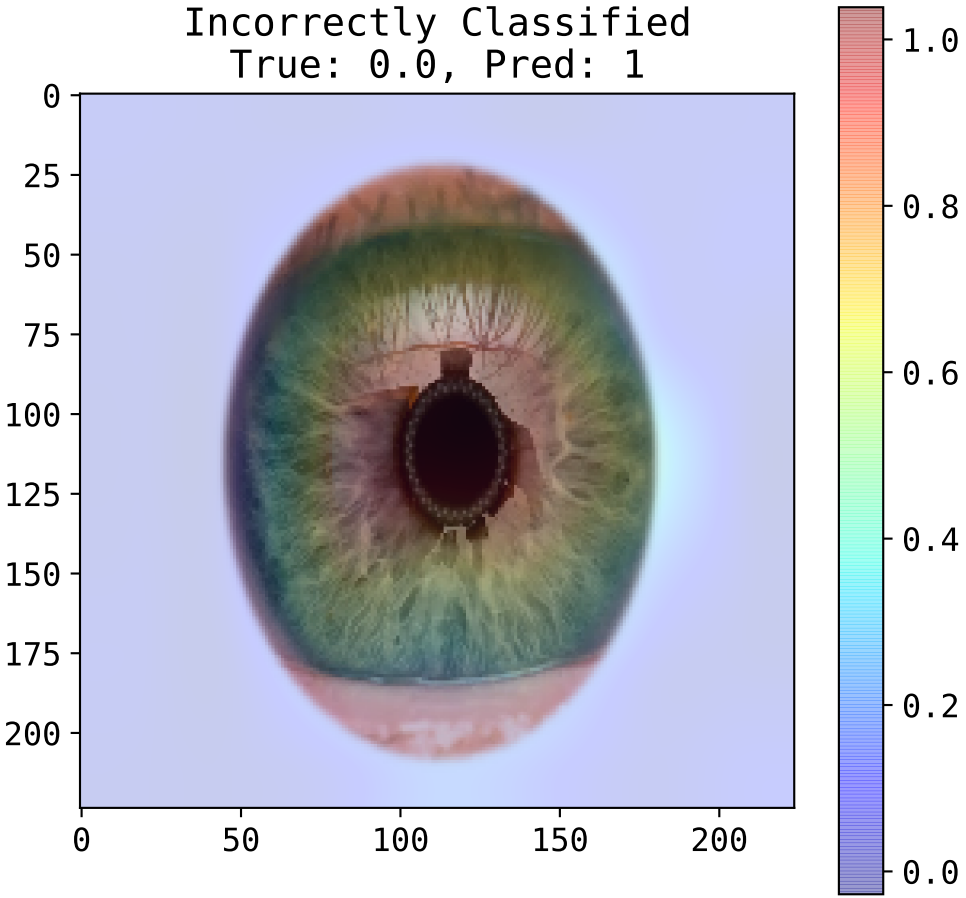

Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Incorrectly Classified
True: 0.0, Pred: 1
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Incorrectly Classified
True: 0.0, Pred: 1
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Incorrectly Classified
True: 0.0, Pred: 1
0
25
50
75
100
125
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150
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0.8
0.6
0.4
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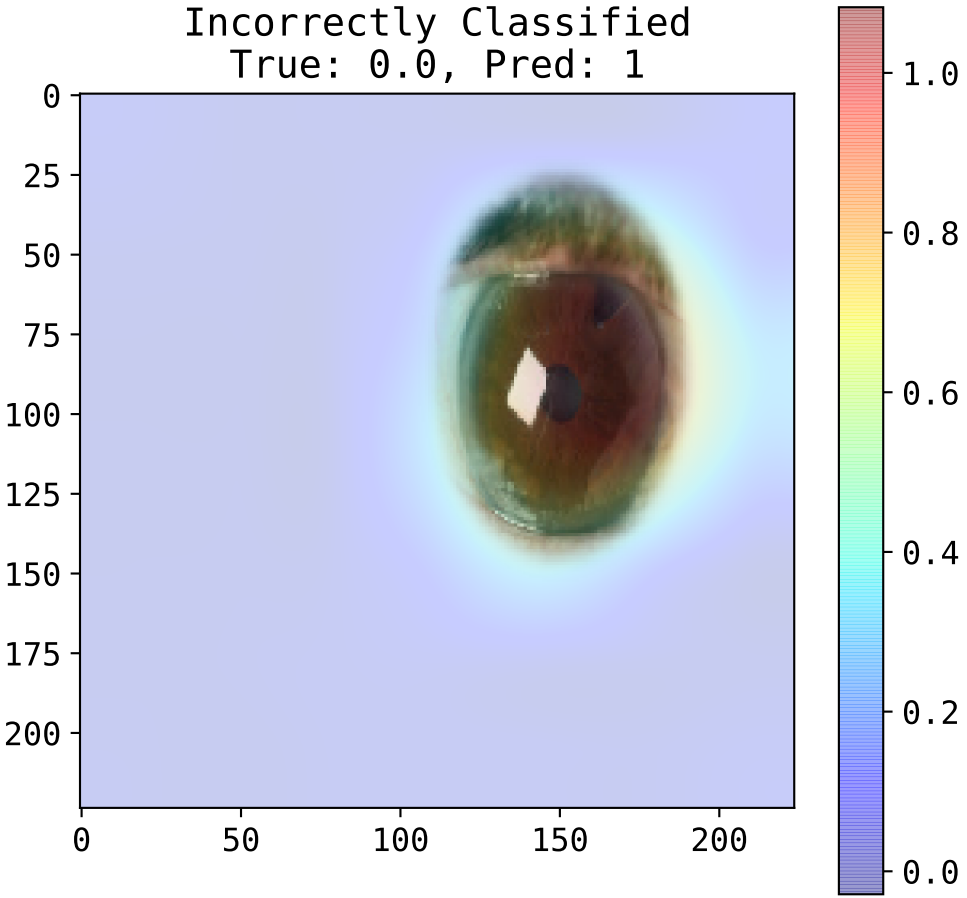
0.0


Incorrectly Classified
True: 0.0, Pred: 1
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150
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1.0
0.8
0.6
0.4
0.2
0.0